Low-voltage power distribution and control systems > Panelboards >

Pow-R-Line 4X panelboards

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Powering Business Worldwide

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Lighting and Distribution Panelboards Overview

- Designed for sequence phase connection of branch circuit devices
- Enables single-, two- or three-pole arrangements for a balanced electrical load on each phase
- Sturdy, rigid chassis assembly ensures accurate alignment of interior with panel front
- Assembly also prevents flexing and minimizes loosening or damage to current-carrying parts
- Four-point in-and-out adjustment of panel interior meets critical depth dimensions on flush installations
- Main lugs are approved for copper or aluminum conductors

Available Ratings

- Rated at 240 Vac, 480 Vac and 600 Vac
- Fault current available up to 200 kAIC at 240 Vac, 100 kAIC at 480 Vac and 65 kAIC at 600 Vac

Panelboard Options

- Aluminum, copper, silver-plated copper or tin-plated copper bus
- Aluminum or copper lugs
- Density-rated bus
- Ground bars
- Customer-owned meters
- Service entrance equipment construction
- Surge protective devices
- Seismically qualified panelboards



From left to right: Pow-R-Line 1X, Pow-R-Line 2X, Pow-R-Line 4X and Pow-R-Line 3X panelboards

Enclosures

- Boxes are code-gauge galvanized steel, available unfinished or painted ANSI-61 light gray
- Standard panelboard cabinets are designed for indoor use
- Alternate types are available for special-purpose applications
- All enclosures meet UL® standards and include wiring gutters with proper wire bending space
- Box dimensions shown are inside dimensions—for outside dimensions, add 0.25 inch (6.4 mm)
- Standard panelboard boxes are supplied without knockouts (blank endwalls)



The Three-Piece Trim for Larger Power Distribution Panelboards Provides for Easy Handling and Installation

Fronts

- Made of code-gauge steel with a high-durability ANSI-61 light gray finish
- Branch circuit and small power distribution panelboards include a door and concealed hinges
- Flush-type latch and lock assembly included
- All locks are keyed alike
- Trims available in both surface- and flush-mounted designs

Power Distribution Panelboard Fronts Only

- Utilize a breaker front cover design providing each device with a dedicated bolt-on steel cover
- Individual covers form a single deadfront used with two wiring gutter covers to complete the trim
- A door is not finished but can be provided for an additional charge using a deeper-than-standard box

EZ Box™ and EZ Trim™

Provided standard for:

- Pow-R-Line 1X and Pow-R-Line 2X lighting panelboards
- Pow-R-Line 3X, Pow-R-Line 3E and Pow-R-Line 3XF mid-range panelboards



EZ Box and EZ Trim

Features

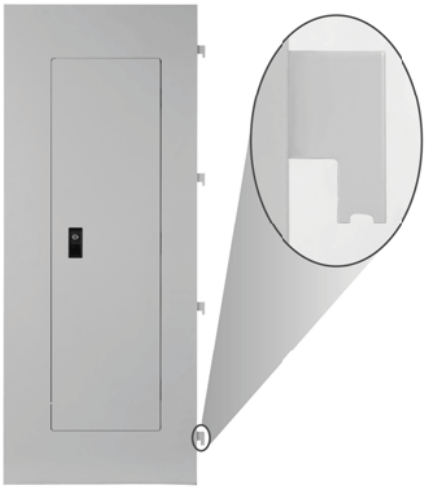
- Virtually eliminates sharp edges
- Trim installs quickly
- Door-in-door is standard
- Ability to adjust flush box to wall irregularities
- Trim installs without the need for tools
- No exposed hardware (because there is none)

EZ Box

- Flanges are bent and painted, and without sharp edges
- All steel panelboard chassis parts are painted to reduce injuries during handling and installation
- Each flange adjusts outward up to 0.75 inch (19.1 mm), enabling installer to flush the box with a wall after wall material is installed
- Flanges also provide means for attaching EZ Trim

EZ Trim

- Design installs in seconds
- Standard trim includes door-in-door construction and no exposed hardware
- No tools required for installation
- Includes hangers attached on the right side
- Bottom trim hanger with notch in base inserts into the bottom right side box flange opening, resting the notch on the flange



Standalone Trim and Bottom Flange Hanger with Notch

Standards and Certifications

When used with Eaton's panelboard chassis, EZ Box and EZ Trim meet the following applicable industry standards:

- UL 50 Listed
- NEMA® Standard PB1
- Federal specifications
- National Electrical Code® (NEC®)

UL Listed

Application Considerations and Definitions

Standards

All Eaton panelboards meet the following applicable industry standards, except where noted:

- UL
  - Panelboards: UL 67
  - Cabinets, boxes and trims: UL 50

Note: Only panelboards containing UL Listed devices can be UL labeled.

- NEC
- NEMA PB 1
- Federal Specification W-P-115c
  - Circuit breaker—Type I Class 1
  - Fusible switch—Type II Class 1

Panelboard Selection Factors

In selecting a panelboard, the following factors must be considered:

- Service (voltage and frequency)
- Interrupting capacity (fully or series rated)
- Ampere rating of main
- Ampere ratings of branches
- Installation environment
- Codes and standards mandates

Panelboard Short-Circuit Rating

- Eaton's assembled panelboards are test verified by, and listed with, UL
- Ratings are generally that of the lowest interrupting rated device in the panel
- Where branch devices have been UL tested in combination with specific main devices having a higher interrupting rating, the series short-circuit rating of the assembled panelboard will be the same as the series tested rating of the approved rated main breaker
- Series ratings apply to panels having main devices, or main-lug-only panelboards fed remotely by the device listed in the series ratings chart as the main, for which UL Listed tests were conducted

Selective Coordination

Refer to Molded Case Circuit Breakers Design Guides for information on overcurrent protective device combinations used in selectively coordinated systems.

Service Entrance Equipment

Service Entrance Equipment NEC Articles 230.62 and 230.71, and UL, require that:

- Panelboards used as service entrance equipment must be located near the point where the supply conductors enter the building
- A single service disconnect within the panelboard
- Must include service disconnect barrier, connector for bonding and grounding neutral conductor
- A service-entrance-type UL label must be factory installed
- Ground fault protection of equipment shall be provided for solidly grounded wye electrical services of more than 150 V to ground, but not exceeding 600 V phase-to-phase for each service disconnecting means rated 1000 A or more
- Service entrance panels must be identified as such on the order entry to the manufacturing location

Table 22.3-1. Service Entrance Main Breaker Kits

Description	Circuit Breaker Frame	Catalog Number
Service entrance barrier for PD5	PD5	**PRLSEBPD5**
Service entrance barrier for PD4	PD4	**PRLSEBPD4**
Service entrance barrier for PD3	PD3	**PRLSEBPD3**
Service entrance barrier for PD2	PD2	**PRLSEBPD2**
Service entrance barrier for GHB	GHB	**PRLSEBGHB**
Service entrance barrier for BAB	BAB	**PRLSEBBAB**

Column Type Panelboards

The same general code restrictions apply as for standard-width panels except where trough extensions are used.

Multi-Section Panelboards

- When more than 42 overcurrent protective devices are required, two or more separate enclosures may be required
- Separate fronts for each box are standard

Interconnecting Multi-Section Panelboards

- When a panelboard must be furnished in more than one section (box) for connection to one feeder, each section must be furnished with main bus and terminals of the same rating unless a main overcurrent device is provided in each section
- Sub-feed or through-feed provisions must also be added to provide connection capability to the second section

Note: Sub-feed or through-feed lugs cannot be used on any panelboard not protected by a single main overcurrent device either in the panelboard or immediately upstream, i.e., service entrance panelboards with main-lug-only using the six disconnect rule.

Sub-Feed Lugs (Figure 22.3-1)

- Sub-feed (second set of) lugs interconnect multi-section panels
- Mounted directly beside the main lugs
- Required in each section except the last panel in the lineup
- Feeder cables are brought into the wiring gutter of the first section and connected to the main lugs
- Another set of same-size cables connect to the sub-feed lugs (Section 1) and are carried over to the main lugs of the adjacent panel
- Cross connection cables are not furnished by Eaton
- Sub-feed lugs are only available on main-lug-only panels

Note: Sub-feed lugs may not be used on main-lug-only (six disconnect rule) service entrance panels.

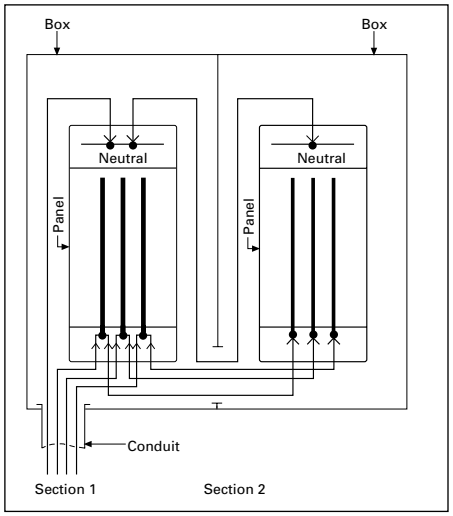


Figure 22.3-1. Sub-Feed Lugs

Through-Feed Lugs (Figure 22.3-2)

- Another method to interconnect multi-section panelboards
- Incoming feeder cables connect to main lugs or main breaker at the bottom of panel (Section 1)
- Another set of lugs (through-feed) are located at the opposite end of the main bus
- Interconnecting cables connect to the through-feed lugs in Section 1 and are carried over to the main lugs in Section 2
- Connection arrangement is reversible, with main lugs at top and through-feed lugs at bottom of end of panel
- Cross connection cables are not furnished by Eaton

Note: Through-feed lugs may not be used on main-lug-only (six disconnect rule) service entrance panels.

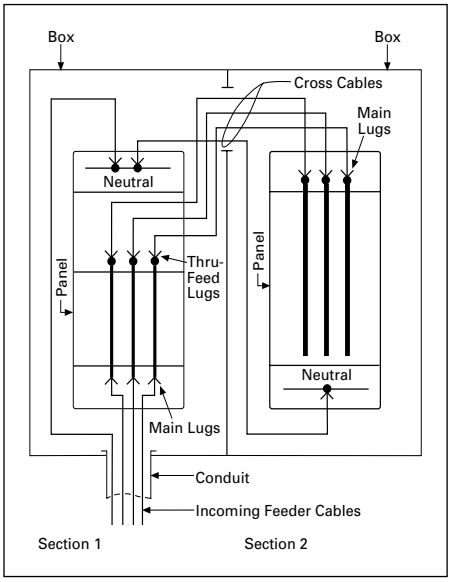


Figure 22.3-2. Through-Feed Lugs

Multiple Section Panelboard—Flush Mounted

Shown below is the standard method for flush mounting multiple section lighting and distribution panelboards using standard flush trims.

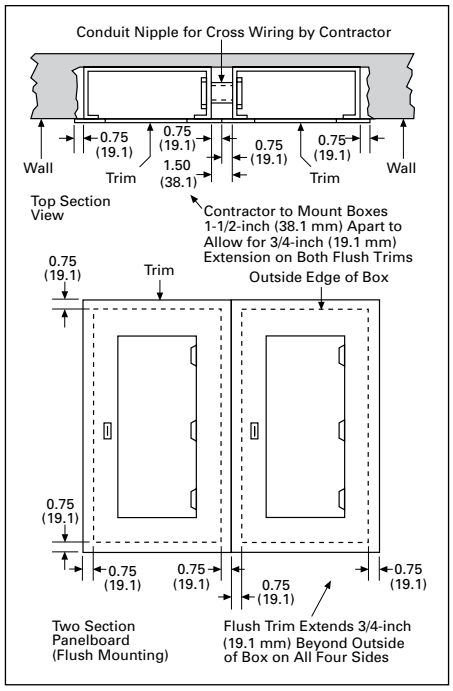


Figure 22.3-3. Multiple Section Panelboard—Flush Mounted—Dimensions in Inches (mm)

Branch Circuit Loading for Lighting Panels

The size of mains and branches should be selected based on the following:

- Lighting circuits: NEC Article 210, 215, 220 and 240
- Distribution circuits, actual or continuous loads: NEC Article 384.16
- Motor circuits: NEC Article 430
- Diversity factor
- Provision for future loading

Overcurrent Protection

- NEC Article 408 states a panelboard shall be protected by an overcurrent protective device having a rating not greater than that of the panelboard
- The overcurrent protective device shall be located within, or at any point on the supply side of, the panelboard
- Exceptions to Article 408 selectively apply—refer to it for specifics

Ground Fault Protection

May be added to most panelboards using Eaton's integral molded case circuit breaker and included feeder devices on:

- Power panelboards
- Mains on all panelboards

Arcflash Reduction Maintenance System™

- Available on many molded case circuit breakers from 70 A to air power circuit breakers at 5000 A
- Enables breakers to trip quickly, significantly reducing the available arc flash potential
- Recognized by the 2011 NEC and the National Electrical Safety Code® (NFPA 70E®)

Combination AFCI Circuit Breakers

- Eaton's 125 Vac AFCI single- and two-pole, 15 A and 20 A bolt-on breakers in panelboards meet NEC Article 210.12
- See the NEC for definitions and details

Ambient Temperatures

The primary function of an overcurrent device is to protect the conductor and its insulation against overheating. In selecting the size of the devices and conductors, consideration should be given to the ambient temperature surrounding the conductors within and external to the panelboard. Cumulative heating within the panelboard may cause premature operation of overcurrent protective devices.

- UL test procedures are based, in part, on 80% loading of panelboard branch circuit devices
- NEC Article 408 limits the loading of overcurrent devices in panelboards to 80% of rating where in normal operation, the load will continue for three hours or more
- Further derating may be required, depending on ambient temperature, duty cycle, frequency or altitude
- An exception applies to assemblies and overcurrent devices approved for continuous duty at 100% of their rating and is covered in NEC 210.20 (a)—for additional information, see Molded Case Circuit Breakers Design Guides

Special Conditions

Standard panelboards, assembled with standard components, are adequate for most applications. However, special consideration should be given to those required for application under special conditions, such as:

- Excessive vibration or shock
- Frequencies above 60 cycles
- Altitudes above 6600 feet (2012 m)
- Damp environment (possible fungus growth)
- Compliance with federal, state and municipal electrical codes and standards

Seismic Qualification

Refer to Power Distribution Systems Design Guides for information on seismic qualification for this and other Eaton products.

Harmonic Currents

- Standard panelboard neutrals are rated for 100% of the panelboard current
- Because harmonic currents can cause overheated neutrals, an option is provided for neutrals to be rated at 200% (1200 A maximum neutral for 600 A main bus) of the panelboard phase current
- Panelboards with the 200% rated neutral are UL Listed as suitable for use with nonlinear loads
- Prior to specifying the 200% rated neutral, a harmonic survey of the distribution system is recommended

Surge Protective Devices (SPD)

The quality of power feeding sensitive electronic loads is critical to the reliable operation of any modern facility. The most frequent causes of microprocessor-based equipment downtime and damage are voltage transients and electrical noise.

- High-energy transients include lightning-induced surges and power company switching that can destroy components instantly
- Continual low-energy transients and high-frequency noise, which are more frequent, can cause erratic equipment performance or sudden failure of electronic circuit board components

Eaton's SPD provides protective and diagnostic systems that are integral to panelboards:

- Safeguards sensitive electronic equipment from the damaging effects of high- and low-energy transients
- Integrates into the panelboards using a "zero lead length" direct busbar connection
- Uses integral disconnect on all panelboards

For complete product description and available ratings, refer to Surge Protection (SPD) & Power Conditioning Products Design Guides.



Eaton SPDs May be Integrated into Most Panelboards

Compact Panelboard Meter

Most Eaton panelboards can integrate a meter for reading the panelboard power and energy usage. Our Power Xpert™ Meter 350 offers:

- ANSI 12.20 0.5% accuracy
- A bright backlit LCD display
- Real energy pulse output
- Phase loss alarm
- Optional RS-485 communication capability

Pow-R-Line 4X



PRL4X Circuit Breaker Panelboard

General Description

Panelboard Ratings

Voltage

- 240 V, 480 V or 600 Vac maximum
- 600 Vdc maximum

Main Lugs

- 250–1200 A

Main Breakers

- 250–1200 A

Branches

- Breakers 15–1200 A, bolt-on
- Breakers 20–600 A drawout
- Fusible switches 30–1200 A, bolt-on

Short-Circuit Current Ratings (Symmetrical)

- 240 Vac: 10–200 kA fully rated
- 240 Vac: 22–200 kA series rated
- 480 Vac: 14–200 kA fully rated
- 480 Vac: 22–150 kA series rated
- 250 Vdc: 10–22 kA fully rated

Service

- Three-phase, four-wire 208Y/120 V, 240/120 V delta and 480Y/277 V
- Single-phase, three-wire 120/240 V
- Single-phase, two-wire 120 V
- Three-phase, three-wire 120, 240, 480 and 600 V
- Two-wire 125 Vdc
- Two-wire 250 Vdc
- Two-wire 600 Vdc

Suitable for service entrance applications when specified.

Bussing

250–1200 A tin-plated aluminum is standard; copper is available as an option. Density rated bus is also available as an option.

Main Lugs Only

The short-circuit rating of the MLO assembled panelboard will be fully rated based upon the lowest rated branch device or may be series rated with an approved upstream device.

Main lugs only ampere ratings: 250, 400, 600, 800 and 1200.

Main Circuit Breakers

The short-circuit rating shown is that of the main breaker only. The short-circuit rating of the assembled panelboard is the rating of the lowest fully rated main or branch device, or the rating of an approved series rated combination.

Series Rating Combinations

For more information on product series ratings, please refer to the Eaton website at https://www.eaton.com/us/en-us/products/low-voltage-power-distribution-control-systems/series-rating.html or download Eaton Information Manual 1C96944H02.

Pow-R-Line 4DX Drawout Panelboard



Type PRL4DX Drawout Molded Case Circuit Breaker Power Panelboard

General Description

- Drawout molded case circuit breaker power panelboard
- Front accessible
- Front connected
- Through-the-door design drawout mechanism
- Visual indication of breaker status and position
- Large grab handles for easy removal
- 600 Vac maximum
- 1200 A maximum mains
- 600 A maximum drawout molded case feeder breakers

Application Description

- Interrupting ratings up to 200 kAIC symmetrical
- Feeder power panelboard
- Rated as service entrance equipment when appropriately equipped
- Ideal for:
  - ❑ Data centers
  - ❑ Industrial facilities
  - ❑ Process equipment manufacturing
  - ❑ Anywhere that requires quick change of feeder devices is needed

Benefits

- Ease of maintenance
- Faster to remove and install
- Less downtime

Standards and Certifications

- UL 67 Listed chassis
- UL 50 Listed box and trim

Series Rating Combinations

For more information on product series ratings, please refer to the Eaton website at https://www.eaton.com/us/en-us/products/low-voltage-power-distribution-control-systems/series-rating.html or download Eaton Information Manual 1C96944H02.

Circuit Breaker Technical Data

Table 22.3-2. Electrical Characteristics of Circuit Breakers

Circuit Breaker Ratings				UL Listed Interrupting Ratings (kA rms Symmetrical)						
Type	Ampere Rating	Number of Poles	Volts AC	AC Rating, Volts					DC Rating, Volts ①	
				120/240	240	277	480	600	125	250
BAB, HQP	15–70 15–100	1 2	120 120/240	10 10	— —	— —	— —	— —	— —	— —
BAB-H, HQP-H	15–100	2, 3	240	—	10	—	—	—	—	—
BABRP, BABRSP	15–30 15–30	1 2	120 120/240	10 10	— —	— —	— —	— —	— —	— —
QBGF, QPGF, QPGFEP QBGFEP	15–50 15–50	1 2	120 120/240	10 10	— —	— —	— —	— —	— —	— —
QBAF, QBAG	15–20 15–20	1 2	120 120/240	10 10	— —	— —	— —	— —	— —	— —
QBHW, QPHW	15–70 15–100	1 2	120 120/240	22 22	— —	— —	— —	— —	— —	— —
QBHW-H, QPHW-H	15–100	2, 3	240	—	22	—	—	—	—	—
QBHGF, QPHGF, QPHGFEP QBHGFEP	15–30 15–30	1 2	120 120/240	22 22	— —	— —	— —	— —	— —	— —
QBHAF, QBHAG	15–20 15–20	1 2	120 120/240	22 22	— —	— —	— —	— —	— —	— —
GHB	15–100 ② 15–100	1 2, 3	277 480Y/277	65 —	— 65	14 —	— 14 ③	— —	14 —	— 14
GHQ	15–30	1, 2	277	65	—	14	—	—	—	—
HGHB	15–30	1	277	65	—	25	—	—	—	—
GHBGFEP	15–60	1	277	—	—	14	—	—	—	—
GHQRSP	15–100 15–100	1 2	277 480Y/277	65 —	— 65	14 14	— 14 ③	— —	— —	— —
EGB	15–125 15–125	1 2, 3	277 480	35 —	35 35	18 —	— 18	— —	10 —	— 10
EGS	15–125 15–125	1 2, 3	277 480	100 —	— 100	35 —	— 35	— —	35 —	— 35
EGH	15–125 15–125	1 2, 3	277 480	200 —	— 200	65 —	— 65	— —	42 —	— 42
PDG2xF	15–100 15–100	1 2, 3	277 480	— —	— 18	14 —	— 14	— —	10 —	— 10
PDG2xG, PDG3xG ④	15–150 15–225	1 2, 3	277 600	— —	— 65	35 —	— 35	— 18	10 —	— 10
PDD2xF PDD2xG PDD3xG	100–225 100–225 250–400	2, 3 2, 3 2, 3	240 240 240	— — —	22 65 65	— — —	— — —	— — —	10 10 —	— — 10
PDG3xG ④⑤, PDF3xG ⑥ LHH ⑦ NHH PDG3xG ④ CLD ⑤⑥ PDG4xG ⑤, PDF4xG ⑤⑥ CND ⑤⑥	100–400 150–400 150–350 250–600 300–600 300–800 400–1200	2, 3 2, 3 3 3 2, 3 2, 3 2, 3	600 480 600 600 600 600 600	— — — — — — —	65 100 100 65 65 65 65	— — — — — — —	35 65 65 35 35 50 50	25 35 35 18 25 25 25	— — — — — — —	10 42 — 22 22 ⑦ 22 —
CNGC CNGH CNGS PDG5xP PDG5xM NGS	800, 1200 800, 1200 800, 1200 800, 1200 800, 1200 800, 1200	— — — — — —	— — — — — —	— — — — — —	200 100 85 200 100 85	— — — — — —	100 65 50 100 65 50	65 35 25 65 35 25	— — — — — —	— — — — — —

① DC ratings apply to substantially non-inductive circuits.
② DC rated single-pole, 15–70 A only.
③ Rating 480Y/277 Vac maximum.
④ The 400 A frame must use trip units of ratings 100–400, while the 600 A frame must use trip units of ratings 500, 600 or designated by H, such as H250. The H as the leading character of the ampacity indicates a high instantaneous version of the breaker for coordination purposes. H ratings must use 600 A frame.
⑤ Available with integral ground fault protection.
⑥ 100% rated breaker.
⑦ DC rating not available with PXR trip units.

Table 22.3-2. Electrical Characteristics of Circuit Breakers (Continued)

Circuit Breaker Ratings					UL Listed Interrupting Ratings (kA rms Symmetrical)							
Type	Ampere Rating	Number of Poles	Voltage		AC Rating, Volts					DC Rating, Volts ①		
			AC	DC	120/240	240	277	480	600	125	250	600
High Interrupting Capacity Circuit Breakers												
PDG2xM	15–150 15–225	1 2, 3	277 600	— —	— —	— 100	65 —	— 65	— 25	10 —	— 22	— —
PDD2xM PDG3xM ②③, PDF3xM ④ PDG3xM ②③	100–225 100–400 250–600	2, 3 2, 3 2, 3	240 600 600	— — —	— — —	100 100 100	— — —	— 65 65	— 35 35	10 — —	— 22 22	— — —
PDG4xM ③, PDF4xM ③④ PDG5xM ③	300–800 400–1200	2, 3 2, 3	600 600	— —	— —	100 100	— —	65 65	35 35	— —	25 —	— —
Current-Limiting Circuit Breakers												
PDD2xP PDG2xP	100–225 15–225	2, 3 2, 3	240 600	— —	— —	200 200	— —	— 100	— 35	10 —	— 22	— —
PDG3xP ②③ PDG3xP ②⑤ PDG5xP ③	100–400 250–600 400–1200	2, 3 2, 3 2, 3	600 600 600	— — —	— — —	200 200 200	— — —	100 100 100	65 50 65	— — —	22 42 —	— — —
Current Limit-R® Circuit Breakers												
FCL	15–100	2, 3	480	—	—	200	—	150	—	—	—	—
TRI-PAC® Current-Limiting Circuit Breakers												
FB-P NB-P	15–100 300–800	2, 3 2, 3	600 600	— —	— —	200 200	— —	200 200	200 200	— —	⑦ ⑦	— —
Direct Current (DC) Rated Breakers												
HFDDC ⑥⑦ HJDDC ⑥⑧ HKDDC ⑥⑧	15–150 70–250 100–400	2, 3 2, 3 2, 3	— — —	600 600 600	— — —	— — —	— — —	— — —	— — —	42 42 42	42 42 42	35 35 35
HLDDC ⑥⑧ HMDLDC ⑥⑧ NBDC ⑥⑧	300–600 300–800 700–1200	2, 3 2, 3 2, 3	— — —	600 600 600	— — —	— — —	— — —	— — —	— — —	42 42 42	42 42 42	35 35 35

① DC ratings apply to substantially non-inductive circuits.
② The 400 A frame must use trip units of ratings 100–400, while the 600 A frame must use trip units of ratings 500, 600 or designated by H, such as H250. The H as the leading character of the ampacity indicates a high instantaneous version of the breaker for coordination purposes. H ratings must use 600 A frame.
③ Available with integral ground fault protection.
④ 100k based on NEMA test procedure.
⑤ DC rating not available with PXR trip units.
⑥ For use on DC systems only.
⑦ Non-interrupting trip type.
⑧ Interrupting trip type.

Terminal Wire Ranges, Pressure-Type Al/Cu Terminals Except as Noted

Where copper-aluminum terminals are supplied on designated panelboard types, best results are obtained if a suitable joint compound is applied when aluminum conductors are used.

Table 22.3-3. Standard Main Lug Terminals

Panel Type	Wire Size Ranges for Ampere Capacity						
	100 A	225 A	250 A	400 A	600 A	800 A	1200 A
Pow-R-Line 4X	—	—	#4–500 kcmil	(2) #4–500 kcmil	(2) #4–500 kcmil	(3) #4–500 kcmil	(4) #4–500 kcmil

Note: Optional 750 kcmil mechanical screw-type terminals are available upon request. Panelboard dimensions may be affected. Refer to Eaton.

Table 22.3-4. Standard Main Breaker and Branch Breaker Terminals

Breaker Type	Ampere Rating	Wire Size Ranges
BAB, QBHW, BABRSP HQP, QPHW	15–70 90–100	#14–#4 #8–1/0
PDD2xF, PDD2xG PDD2xM, PDD2xP ①	100–225	#4–4/0 or #6–300 kcmil
EGB, EGE, EGS, EGH	15–50 60–125	#14–3/0 Al/Cu #6–3/0 Al/Cu
PDG2xF, PDG2xG PDG2xM, PDG2xP ①, HFDDC ②	15–100 125–225	#14–1/0 #4–4/0
FCL	15–100	#14–1/0
GHB, HGHB, GHQ, GHQRSP	15–50 25–100	#14–1/0 #10–1/0
HJDDC ②	70–250	#4–350 kcmil
PDD3xG	250–350 400	250–500 kcmil (2) 3/0–250 kcmil or (1) 3/0–500 kcmil
PDG3xG ③ PDG3xM ③, PDG3xP ③ HKDDC ②, PDF3xG ① PDF3xM ①	225 350 400	(1) #3–350 kcmil (2) 3/0–250 kcmil or (2) 3/0–250 kcmil or (1) 3/0–500 kcmil
LHH	150–400	#2–500 kcmil (2) #2–500 kcmil or (1) 500–750 kcmil
PDG3xG ③, PDG3xM ③, PDG3xM ③	250–400 500–600	(1) #2–500 kcmil (2) #2–500 kcmil
CLD ①	300–500	(2) 250–350 kcmil
PDG4xG, PDG4xM, HMDLDC ① PDF4xG, PDF4xM	400–600 700–800	(2) #1–500 kcmil (3) 3/0–400 kcmil
PDG5xM, CND PDG5xP, CLD ①	800–1000 1200	(3) 3/0–400 kcmil (4) 4/0–500 kcmil
NGS, PDG5xM, PDG5xP CNGS, CNGH, CNGC	400–1200	(4) 4/0–500 kcmil or (3) 500–750 kcmil

① Suitable for DC applications only.
② LHH is 400 A maximum.
③ The 400 A frame must use trip units of ratings 100–400, while the 600 A frame must use trip units of ratings 500, 600 or designated by H, such as H250. The H as the leading character of the ampacity indicates a high instantaneous version of the breaker for coordination purposes. H ratings must use 600 A frame.

Table 22.3-5. Fusible Switch Terminals

Ampere Rating	Wire Size Ranges
30 60 100	#14–1/0 #14–1/0 #14–1/0
200	#4–300 kcmil
400	250–750 kcmil or (2) 3/0–250 kcmil
600	(2) #4–600 kcmil or (4) 3/0–250 kcmil
800	(3) 250–750 kcmil or (6) 3/0–250 kcmil
1200	(4) 250–750 kcmil or (8) 3/0–250 kcmil

Power Xpert Release Trip Unit for Molded Case Circuit Breakers

Description

Eaton's Power Xpert Release (PXR) trip units are programmable communicating microprocessor-based low-voltage electronic trip unit systems for Eaton molded case circuit breakers. PXR trip units are available in three models: PXR 10, PXR 20 and PXR 25.

Standards and Certifications

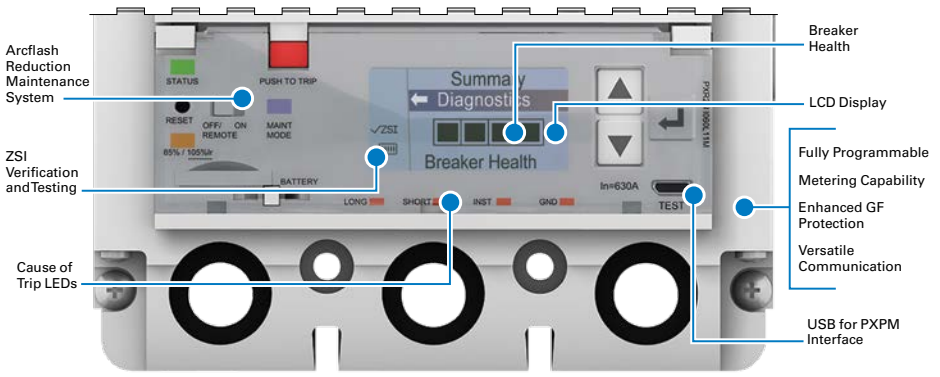
The PXR trip units are listed by Underwriters Laboratories (UL) and Canadian Standards Association (CSA) for use in Power Defense™ Frame 2, 3, 4, 5 and 6 molded case circuit breakers. All PXR units have also passed the IEC 60947-2 test program that includes EMC testing. All trip units meet the low-voltage and EMC directives and carry the CE mark.

Features

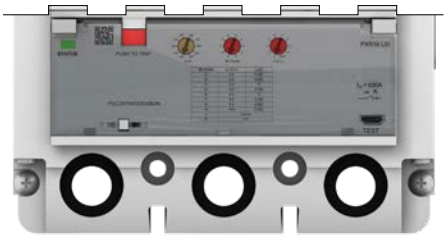
The PXR electronic trip units provide an enhanced and easy-to-use interface that enables end users and maintenance engineers to more easily change set points, test and configure circuit breakers, and review energy and power information. The Power Xpert Protection Manager (PXPM) software also provides the capability of secondary injection tests and reports on-demand without the need of expensive test kits.

Advanced features include:

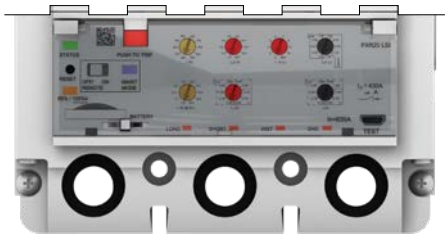
- Industry-first breaker health algorithms provide real-time monitoring and communication of breaker condition
- Cause of trip LED indication and trip event data storage
- Zone selective interlocking (ZSI) verification and testing indication
- Adjustable Arcflash Reduction Maintenance System™ settings
- LCD display with programmable settings



PXR 25 Trip Unit



PXR 10



PXR 20

Table 22.3-6. PXR Features

Features	PXR 10	PXR 20	PXR 25
Protection types	LSI	LSI/LSIG	LSI/LSIG
Status indication	●	●	●
USB secondary injection testing	●	●	●
Programmable by USB port (PXPM)	●	●	●
Independent instantaneous adjustment	●	●	●
Adjustable L, S, I, G pickup and time		●	●
Cause of trip indication	▲	●	●
Load alarm indication with 2 levels		●	●
Programmable load alarm levels			●
Ground fault protection and alarm		○	○
Arcflash Reduction Maintenance System Available PD3, PD4, PD5, PD6 to meet NEC 240.87 requirements		○	○
Zone selective interlocking (ZSI) with indication		○	○
Programmable relays		○	●
Modbus RTU communication		○	●
CAM module communication		○	○
Rotatable LCD display			●
Breaker health and diagnostic monitoring		▲	●
Current metering accuracy		2.0%	0.5%
Voltage metering accuracy			0.5%
Power and energy metering accuracy			1.0%

Legend:

● Standard

○ Optional

▲ Available through USB port (PXPM)

Metering, Monitoring and Surge Protective Devices



Power Xpert Meter 1000



Power Xpert Meter 2000



Power Xpert Meter 3000

Power Xpert 1000 Meter

The Power Xpert Meter 1000 (PXM1000) series power and energy meters monitor the most critical aspects of an electrical distribution system. This premier metering instrument uses advanced technology to make it simple to use, powerful, scalable and highly flexible.

The PXM1000 is a revenue grade power and energy meter that delivers a cost-effective solution for energy and sub-metering applications. This three-phase meter provides high accuracy and advanced features in the standard 4-inch form factor and can be expanded with multiple modular I/O options.

Meter features include:

- ANSI C12.20 and IEC 62053-22 utility billing accuracy will help meet stringent customer specifications
- Available in 5 A and 333 mV CT type inputs, allowing ease of use in multiple applications
- Rogowski coils allow for ease of use in retrofit applications
- Multiple protocols including Modbus TCP and BACnet/IP and with available HTTP push, allowing data to be sent to the cloud to help meet energy code data storage requirements

Power Xpert 2250 Meter

The Power Xpert 2250 meter provides all the core functions for monitoring power consumption and power quality, ethernet connectivity and onboard gateway card limits. This unit uses D/A technology to sample circuits at 400 samples per cycle for extremely accurate measurement of power factor and energy consumption. In addition, the meter has 256 MB for logging meter data.

Power Xpert 2260 Meter

The Power Xpert 2260 meter adds the ability to monitor total harmonic distortion and the ability to set onboard meter limits. The meter also will illuminate LEDs on the faceplate, indicating that a limit has been exceeded and provides 512 MB for data logging.

Power Xpert 2270 Meter

The Power Xpert 2270 meter adds the ability to monitor individual harmonics and visualize waveforms on your desktop using the embedded web server and raises the storage to 768 MB for data logging.

Meter features include:

- Data provided to fully understand your facility's power quality
- Detailed event information; pinpoint the root causes of problems—or prevent them from occurring
- Measure, trend and analyze power via information from onboard web and comma separated values (CSV) exporting capabilities
- Up to 768 MB of storage; typically 15 years of storage capability depending on the meter model and frequency of events
- Local or remote configuration

Power Xpert Meter 3000

The Power Xpert Meter 3000 Series power quality and energy meter monitors the most critical aspects of an electrical distribution system. This premier metering instrument uses advanced technology to make it simple to use, powerful, scalable and highly flexible.

Meter features include:

- ANSI C12.20 and IEC 62053-22 revenue grade accuracy will help meet stringent customer specifications
- Combines state-of-the-art technology with harmonic viewing, data trending, performance benchmarking and sag/swell waveform recording through an embedded web server accessible from any location via a standard web browser
- The embedded web server supports viewing of triggered waveforms one channel at a time, including the ability to zoom and scroll horizontally using a slider bar. Waveforms are stored in the meter's nonvolatile flash memory using an industry standard COMTRADE format, allowing you to view and analyze the waveforms in any standard free-of-charge or commercial COMTRADE file viewer
- Offers maximum flexibility in waveform recording by allowing the number of recorded waveform cycles, number of pre- and post-event cycles, and sampling rate to be configured to your system requirements
- Standard protocol ModbusTCP and BACnet/IP
- Available in 1 A / 5 A current transformer (CT) type inputs for ease of use in traditional installations

For information on other available power meters, visit **www.eaton.com/meters**.

Metering, Monitoring and Surge Protective Devices, Continued



Power Xpert Multi-Point Meter

Power Xpert Multi-Point Meter

Eaton Power Xpert Multi-Point meter (PXMP) helps facility managers track and accurately allocate energy usage among tenants or departments in office buildings, shopping malls, industrial sites, universities, campuses and apartment and condominium complexes. Power Xpert Multi-Point Meters monitor, quantify and help benchmark energy usage.

Meter features include:

- Meets rigid ANSI C12.20 and IEC 62053-22 accuracy specifications for revenue meters
- Quick connect terminals for current sensors, Modbus communication and bus voltages
- Monitors power and energy for up to 60 current sensors; scalable from 6 to 60 circuits
- 256 MB of standard memory for up to two years of 15-minute interval data
- Extensive LEDs for verification of sensor connections, communication status and equipment status
- Automatically detects rating of each current sensor; current sensors are self-protecting in the event of an open circuit condition under load for added safety and reliability





Power Xpert Branch Circuit Monitor

Power Xpert Branch Circuit Monitor

Eaton's Power Xpert Branch Circuit Monitor (PXBCM) provides remote access to live energy readings and facilitates data integration for data center intensive industries, facilities working to optimize server capacity and companies with a critical need to maintain uptime. By combining monitoring capabilities down to the plug level with overload alerts, which indicate when circuits are close to exceeding thresholds, the PXBCM helps minimize or prevent downtime.

- ANSI C12.20 0.5% revenue accuracy for panelboard applications
- Satisfies energy code requirements to meter energy by branch circuit and report energy consumption by load group
- Modular form factor supports either pre-mounted CTs (PXBCM-MMS) or loose CTs (PXBCM-MME) and makes it easy to scale for 18-, 30-, 42- and 84-circuit panelboard applications



Power Xpert Gateway

Power Xpert Gateway

Eaton's Power Xpert Gateway (PXG) bridges the IT and facilities management worlds by bringing disparate panelboards, other power equipment onto the network. The PXG takes the complexity out of connecting power equipment to the network. The out-of-the-box device can support up to 96 devices, translate most industrial communication protocols, and offer user-selectable events and real-time trending. It also features email notification of events, waveform capture and data/event logging—all with no special software. Adding basic meters or the utility's meter, the PXG assists in tracking energy usage. The PXG recognizes the interdependence of IT systems and power systems, and delivers what organizations need to bring these worlds together for seamless, end-to-end system reliability.

The PXG consolidates data available breakers, meters, motor controllers and protective relays, and presents the information in a variety of ways (a web browser being the most widely used method). The PXG is a stand-alone solution, but as needs change and grow, the PXG can be integrated through Power Xpert Software into a broader solution that encompasses other intelligent hardware and can integrate with third-party network management systems (NMS) or building management systems (BMS) for system-wide monitoring and reporting of power and IT.

For more information, visit **www.eaton.com/meters**.

Metering, Monitoring and Surge Protective Devices, Continued



Power Distribution Monitoring and Control

Power Distribution Monitoring and Control

Power Distribution Monitoring and Control helps keep your critical facilities up and running by enabling interaction with low-voltage panelboards and switchboards through intelligent component control, communications and monitoring. Use of Power Distribution Monitoring and Control may simplify installation and maintenance efforts while also reducing exposure to arc flash and providing access to aggregate device data from electronic meters and Eaton's Power Xpert Release (PXR) trip units installed in our globally rated, Power Defense molded case circuit breakers (MCCBs).

For more information, visit **www.eaton.com/meters**.



Integrated Surge Protective Devices

Integrated Surge Protective Devices

Eaton integrates our industry-leading surge protective devices (SPD) into panelboards. Lead length is kept to a minimum to maximize SPD performance. SPD units are available with ratings up through 400k, and are UL listed and labeled to UL 1449 3rd Edition.

All panelboards with integrated SPD units are connected to a lineside overcurrent protective device for disconnecting means. When applied on the lineside of a service entrance main, the disconnecting means does not count as a service disconnect per National Electrical Code Article 230.71[A].

For complete SPD product description, application and ratings, visit **www.eaton.com/spd**.

Table 22.3-7. Side-by-Side Comparison of the SPD Series' Available Feature Packages

Feature package comparison	Basic	Standard	Standard with Surge Counter	Power Xpert SPD
Surge protection using thermally protected MOV technology	■	■	■	■
Dual-colored protection status indicators for each phase	■	■	■	■
Dual-colored protection status indicators for the neutral-ground protection mode ①	■	■	■	
Tri-colored protection status indicators for each phase and the neutral-ground protection mode ①				■
Audible alarm with silence button		■	■	■
Form C relay contact		■	■	■
EMI/RFI filtering, providing up to 50 dB of noise attenuation from 10 kHz to 100 MHz		■	■	■
Surge counter with reset button			■	■
Percentage protection remaining status				■
RJ45 Ethernet port for LAN connection, Modbus TCP/IP or BACnet/IP				■
UI webpage and programmable settings				■
Time-and-date stamped surge log and surge categorization				■

① Neutral-ground protection mode available in applicable voltage configurations only.

Pow-R-Line Metering and Monitoring Options

Device	 Pow-R-Line 1X	 Pow-R-Line 2X	 Pow-R-Line 3E	 Pow-R-Line 3X	  Pow-R-Line 4X / 4F
Power Defense Molded Case Circuit Breakers with Power Xpert Release trip units	■	■	■	■	■
Power Xpert Meter 350	■	■	■	■	
Veris E30 Branch Circuit Monitor	■	■	■		
Power Xpert Branch Circuit Monitor	■	■	■	■	■ (Type PRL4X only)
Power Xpert Meter 1000					■
Power Xpert Meter 2000					■
Power Xpert Meter 3000					■
Power Xpert Gateway					■
Power Distribution Monitoring and Control					■

PRL4X Layout Guide

Technical Data and Specifications

Table 22.3-8. Main Circuit Breakers—Type PRL4X and PRL4DX

Breaker Frame Amperes	Breaker Type	Interrupting Rating (kA Symmetrical)				
		240 Vac	480 Vac	600 Vac	250 Vdc	600 Vdc
250	JD	65	35	18	10	—
250	HJD	100	65	25	22	—
250	JDC	200	100	35	22	—
250 ①	HJDDC	—	—	—	42	35
350	NHH	100	65	35	—	—
400	PDD3xG	65	—	—	10	—
400	PDG3xG ②④	65	35	25	10	—
400	PDF3xG ②③	65	35	25	10	—
400	LHH	100	65	35	42	—
400	PDG3xM ②④	100	65	35	22	—
400	PDF3xM ②③	100	65	35	22	—
400	PDG3xP ②④	200	100	65	22	—
400 ①	HKDDC	—	—	—	42	65
600	PDG3xG ④⑤	65	35	18	22	—
600	PDG3xM ④⑤	100	65	35	22	—
600	PDG3xP ④⑤	200	100	50	42	—
600	CLD ②③	65	35	25	22	—
800	PDG4xG ②	65	50	25	22	—
800	PDF4xG ②③	65	50	25	22	—
800	PDG4xM ②	100	65	35	25	—
800	PDF4xM ②③	100	65	35	25	—
800	NB-P	200	200	200	⑥	—
800 ①	HMDLDC	—	—	—	42	35
800	NGS	85	50	25	—	—
800	CNGS	85	50	25	—	—
800	PDG5xM	100	65	35	—	—
800	CNGH	100	65	35	—	—
800	PDG5xP	200	100	65	—	—
800	CNGC	200	100	65	—	—
1200	NGS	85	50	25	—	—
1200	CNGS	85	50	25	—	—
1200	PDG5xM	100	65	35	—	—
1200	CNGH	100	65	35	—	—
1200	PDG5xP	200	100	65	—	—
1200	CNGC	200	100	65	—	—

① For use on DC systems only.
② Available with integral ground fault protection.
③ 100% rated circuit breaker.
④ The 400 A frame must use trip units of ratings 100–400, while the 600 A frame must use trip units of ratings 500, 600 or designated by H, such as H250. The H as the leading character of the ampacity indicates a high instantaneous version of the breaker for coordination purposes. H ratings must use 600 A frame.
⑤ Available in drawout configuration for PRL4DX.
⑥ 100,000 AIC based on NEMA test procedure.

Main Fusible Switches

The short-circuit rating shown is that of the main switch only. The short-circuit rating of the assembled panelboards is the rating of the lowest fully rated main or branch device or the rating of an approved series rated combination. (Fuses are not included.)

400 A and 600 A switches with shunt trip will be rated 100 kA.

Note: Circuit breaker panelboards are designated PRL4X. Fusible switch panelboards are designated PRL4F.

Table 22.3-9. Main Fusible Switches

Switch Rating Amperes	Fuse Class	Interrupting Rating (kA Symmetrical)		
		240 Vac	600 Vac	250 Vdc
Switches Rated 240 Vac, 250 Vdc				
200	R,T	200	—	10
400	R,T	200	—	10
600	R,T	200	—	—
800	L	200	—	—
1200	L	200	—	—
Switches Rated 600 Vac				
200	R, J,T	200	200	—
400	R, J,T	200	200	—
600	R, J,T	200	200	—
800	L	200	200	—
1200	L	200	200	—

Table 22.3-10. Branch Circuit Breakers

Breaker Type	Ampere Rating	Number of Poles	Interrupting Rating (kA Symmetrical)							
			120 Vac	120/240 Vac	240 Vac	480 Vac	600 Vac	125 Vdc	250 Vdc	600 Vdc
BAB	15–70	1	10	—	—	—	—	—	—	—
BAB	15–100	2	—	10	—	—	—	—	—	—
BAB	15–100	2, 3	—	—	10	—	—	—	—	—
QBGF, QBGFEP	15–50 ①	1, 2	10	10	—	—	—	—	—	—
QBHW	15–70	1	22	—	—	—	—	—	—	—
QBHW	15–100	2	—	22	—	—	—	—	—	—
QBHW	15–100	2, 3	—	—	22	—	—	—	—	—
QBHGF, QBHGFEP	15–30	1, 2	22	22	—	—	—	—	—	—
GHB ②③	15–100	1	—	—	65	14	—	14	—	—
GHB ③	15–100	2, 3	—	—	65	14	—	—	14	—
GHQ ②③	15–30	1, 2	—	—	65	14	—	—	—	—
HGHB ②③	15–30	1	—	—	65	25	—	—	—	—
GHBGFEP	15–60	1	—	—	—	14	—	—	—	—
PDG2xF	15–100	1	—	—	14	14	—	10	—	—
PDG2xF	15–100	2, 3	—	—	18	14	—	—	10	—
PDG2xG ②	15–100	1	—	—	65	35	—	10	—	—
PDG2xG, PDG3xG ④⑧	15–225	2, 3	—	—	65	35	18	—	10	—
PDG2xM ②	15–100	1	—	—	65	65	—	10	—	—
PDG2xM	15–225	2, 3	—	—	100	65	25	—	22	—
HFDDC ⑤	15–150	2,3	—	—	—	—	—	42	42	35
PDG2xP	15–225	2, 3	—	—	200	100	35	—	22	—
FB-P	15–100	2, 3	—	—	200	200	200	—	⑥	—
PDD2xF	100–225	2, 3	—	—	22	—	—	10	—	—
PDD2xG	100–225	2, 3	—	—	65	—	—	10	—	—
PDD2xM	100–225	2, 3	—	—	100	—	—	10	—	—
PDD2xP	100–225	2, 3	—	—	200	—	—	10	—	—
HJDDC ⑤	70–250	2, 3	—	—	—	—	—	42	42	35
NHH	150–350	3	—	100	—	65	35	—	—	—
PDD3xG	250–400	2, 3	—	—	65	—	—	—	10	—
PDG3xG ⑧	100–400	2, 3	—	—	65	35	25	—	10	—
PDF3xG ⑦	100–400	3	—	—	100	65	35	—	22	—
PDG3xM ⑧	100–400	2, 3	—	—	100	65	35	—	22	—
LHH ⑥	125–400	2, 3	—	—	100	65	35	—	42	—
PDF3xM ⑦	100–400	3	—	—	100	65	35	—	22	—
PDG3xP ⑧	100–400	2, 3	—	—	200	100	65	—	22	—
HKDDC ⑤	100–400	2, 3	—	—	—	—	—	42	42	35
PDG3xG ⑧	250–600	3	—	—	65	35	18	—	22	—
PDG3xM ⑧	250–600	3	—	—	100	65	35	—	22	—
PDG3xP ⑧	250–600	2, 3	—	—	200	100	50	—	42	—
CLD ⑥	300–600	3	—	—	65	35	25	—	22	—

① 50 A is two-pole only.
② Single-pole breaker rated 277 Vac.
③ At 480 V, use on 480Y/277 Vac system only.
④ Three-pole only.
⑤ For use on DC systems only.
⑥ 100,000 AIC based on NEMA test procedure.
⑦ 100% rated breaker. Requires copper bus. K- and N-Frame breakers require density rated copper bus. Not available in Type 12, 4 and 4X enclosures.
⑧ The 400 A frame must use trip units of ratings 100–400, while the 600 A frame must use trip units of ratings 500, 600 or designated by H, such as H250. The H as the leading character of the ampacity indicates a high instantaneous version of the breaker for coordination purposes. H ratings must use 600 A frame.

Table 22.3-10. Branch Circuit Breakers (Continued)

Breaker Type	Ampere Rating	Number of Poles	Interrupting Rating (kA Symmetrical)							
			120 Vac	120/240 Vac	240 Vac	480 Vac	600 Vac	125 Vdc	250 Vdc	600 Vdc
PDG4xG	300–800	2, 3	—	—	65	50	25	—	22	—
PDF4xG ①	300–800	3	—	—	65	50	25	—	22	—
PDG4xM	300–800	2, 3	—	—	100	65	35	—	25	—
PDF4xM	300–800	3	—	—	100	65	35	—	25	—
HMDLDC ②	300–800	2, 3	—	—	—	—	—	42	42	35
NGS	320–800	2, 3	—	—	85	50	25	—	—	—
CNGS ②	320–800	2, 3	—	—	85	50	25	—	—	—
PDG5xM	320–800	2, 3	—	—	100	65	35	—	—	—
CNGH ②	320–800	2, 3	—	—	100	65	35	—	—	—
PDG5xP	320–800	2, 3	—	—	200	100	65	—	—	—
CNGC ②	320–800	2, 3	—	—	200	100	65	—	—	—
NB-P	400–800	2, 3	—	—	200	200	200	—	—	—
NBDC ②	700–1200	2, 3	—	—	—	—	—	42	42	50
NGS	500–1200	2, 3	—	—	85	50	25	—	—	—
CNGS ②	500–1200	2, 3	—	—	85	50	25	—	—	—
PDG5xM	500–1200	2, 3	—	—	100	65	35	—	—	—
CNGH ②	500–1200	2, 3	—	—	100	65	35	—	—	—
PDG5xP	500–1200	2, 3	—	—	200	100	65	—	—	—
CNGC ②	500–1200	2, 3	—	—	200	100	65	—	—	—

① 100,000 AIC based on NEMA test procedure.
② For use on DC systems only.

Table 22.3-11. Branch Fusible Switches (Fuses are not included)

Switch Rating Amperes	Mounting	Fuse Class	Interrupting Rating (kA Symmetrical)		
			240 V	600 V	250 Vdc
Switches Rated 240 Vac, 250 Vdc					
30/30	Twin	R	200	—	10
60/60	Twin	R	200	—	10
100/100	Twin	R	200	—	10
200/200	Twin	R,T	200	—	10
100	Single	R	200	—	10
200	Single	R,T	200	—	10
400	Single	R,T	200	—	10
600	Single	R,T	200	—	—
800	Single	L	200	—	—
1200	Single	L	200	—	—
Switches Rated 600 Vac					
30/30	Twin	R, J	200	200	—
60/60	Twin	R, J	200	200	—
100/100	Twin	R, J	200	200	—
200/200	Twin	J,T	200	200	—
100	Single	R, J	200	200	—
200	Single	R, J,T	200	200	—
400	Single	R, J,T	200	200	—
600	Single	R, J,T	200	200	—
800	Single	L	200	200	—
1200	Single	L	200	200	—

Note: Twin branch switches of different ampere ratings are available, i.e., 30/60, 30/100, 60/100.

Modifications

Enclosures

Types 12, 3R, 4/4X.

Ground Bar

Standard bolted in box with (3) #6–300 kcmil terminals. Aluminum is standard, copper is available as an option.

Trims

Trim with door is available as an option forType 1 enclosures.

Surge Protective Device (SPD)

Integrated onto panelboard chassis. For complete product description and available ratings, refer to Surge Protection (SPD) and Power Conditioning Products Design Guides.

Main Lug (MLO), Main Breaker, Neutral, Through-Feed (TFL) and Sub-Feed Lug (SFL) "X" Space Requirements

(For compression lugs, or other configurations not shown, contact an Eaton sales representative.)

● = Blank means no bus under cover to meet NEC cable bending space.

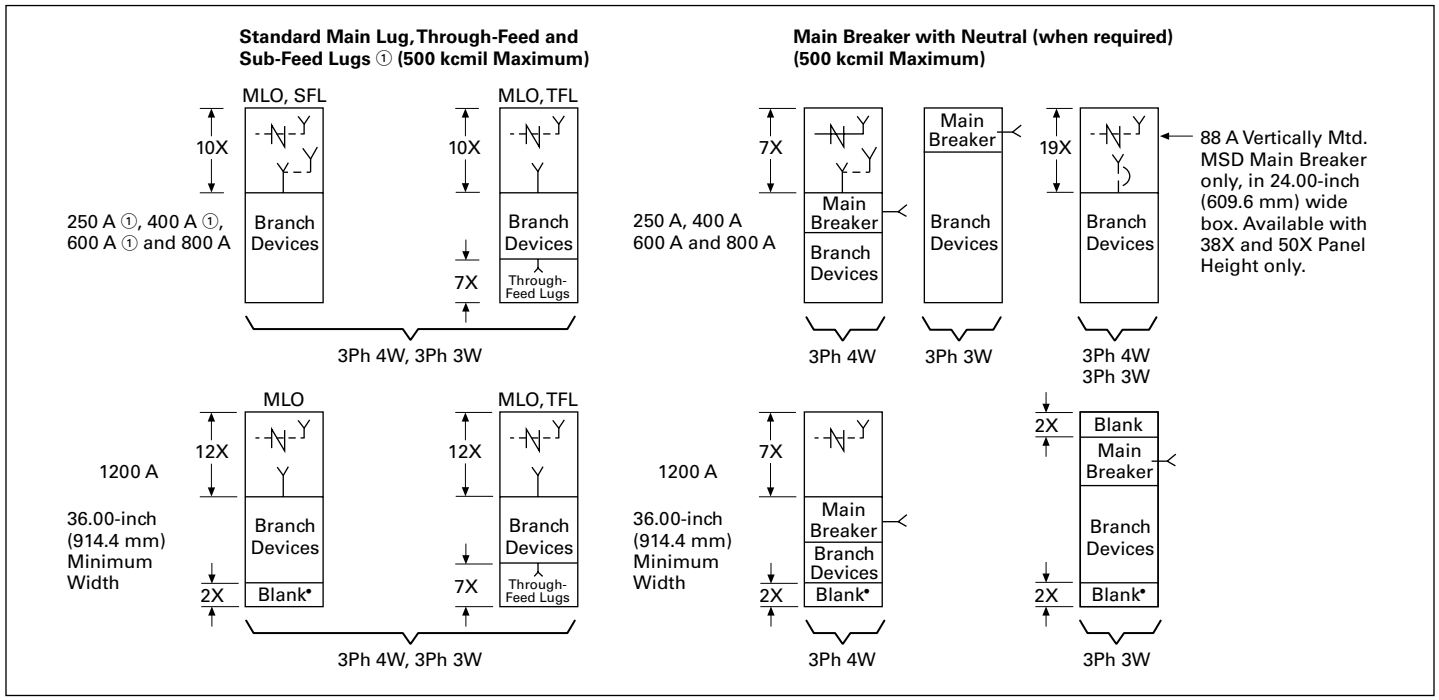


Figure 22.3-4. "X" Space Requirements—Dimensions in Inches (mm)

① Sub-feed lugs are available 250–600 A. For 600 A, use 1200 A "X" space.

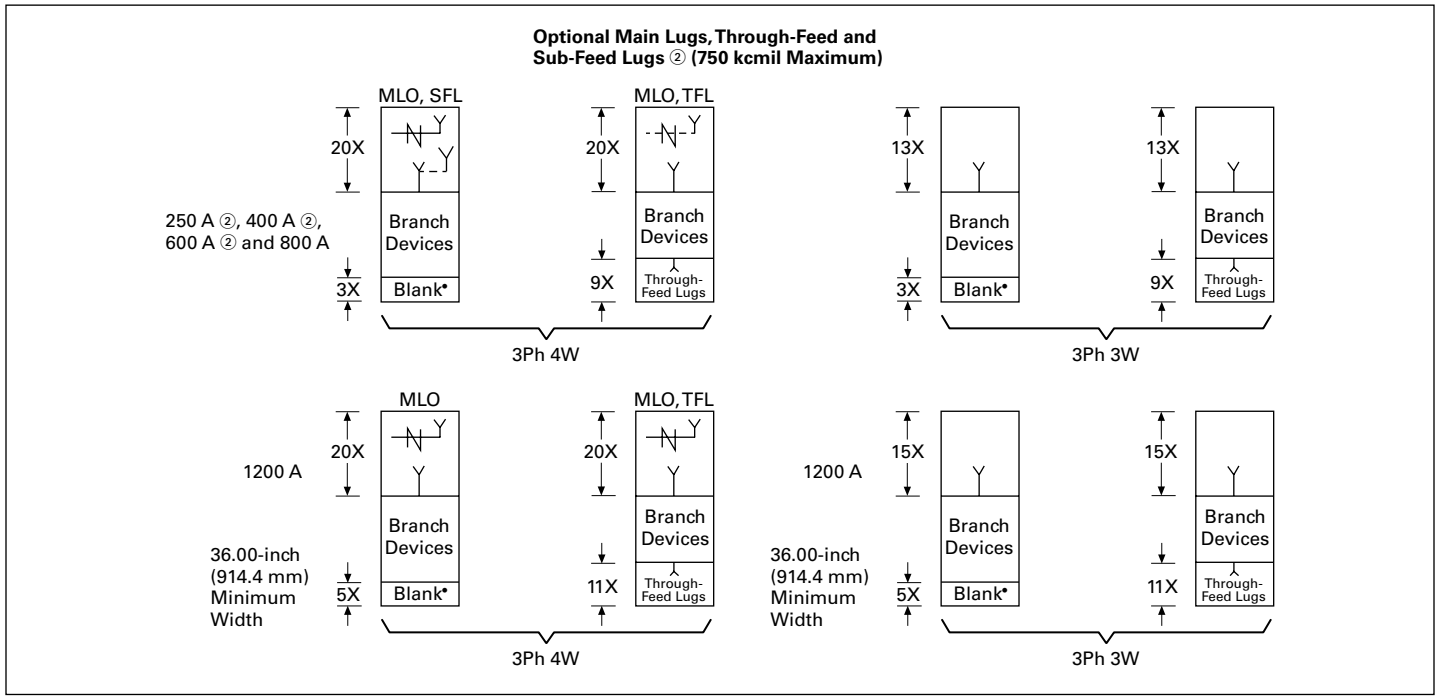


Figure 22.3-5. "X" Space Requirements—Dimensions in Inches (mm)

② Sub-feed lugs are available 250–600 A. For 600 A, use 1200 A "X" space.

Breaker (PRL4X) Type Distribution Panelboards 600 Vac, 250 Vdc

Panelboard Layout and Dimensions

To determine the dimensions of a given panelboard enclosure, make a layout sketch by fitting together the main branch and lug modules according to the appropriate tables in the layout guide. Assign "X" units to each module as shown and obtain a total "X" number.

The height of the enclosure is related to the total "X" units in the layout as shown in **Table 22.3-12**. Three standard box heights are available to accommodate any and all layout arrangements. "X" unit totals that do not exactly match those in **Table 22.3-12** must be rounded off to the next higher standard (26X, 38X, 50X).

When a calculated "X" total for a panel exceeds 50X, the panel must be split into two or more separate sections with "X" space for through-feed lugs figured in for all but one section. If a neutral is required, a separate neutral bar and appropriate "X" space must be included in each section.

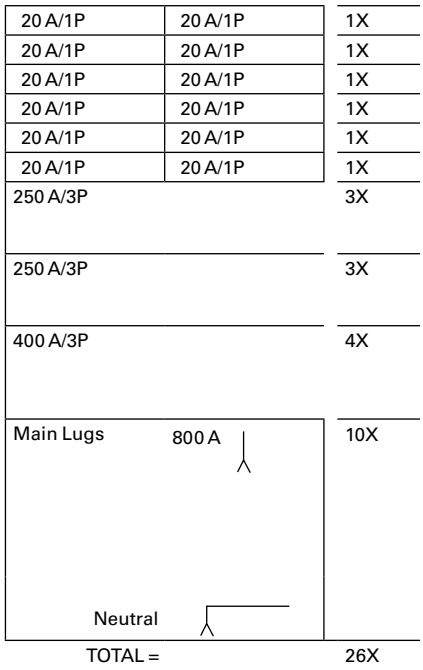
Layout Example

1 – PRL4X panelboard, 480Y/277 V, three-phase, four-wire, 65 kA, 800 A, main lug, consisting of:

12 – 20 A/single-pole HFD

2 – 250 A/three-pole HJD

1 – 400 A/three-pole HKD



Note: For SPD unit, add 7X up to 200 kA and 10X for SPD above 200 kA.

1. From layout guide, total "X" height of panel = 26X, (which is a design standard and no rounding off is necessary).
2. From **Table 22.3-12**, enclosure height for 26X panel = 57.00 inches (1447.8 mm).
3. Width = 24.00 inches (609.6 mm)—directly from layout guide.
4. Total enclosure depth = 11.30 inches (287.0 mm)—standard for all PRL4 panelboards.

Table 22.3-12. Standard Panelboard and Box—Dimensions in Inches (mm)

Panel Height	Box Height	Box Width	Box ① Depth
26X	57.00 (1447.8)	24.00 (609.6) ②	10.40 (264.2)
38X	73.50 (1866.9)	24.00 (609.6) ②	10.40 (264.2)
50X	90.00 (2286.0)	24.00 (609.6) ②	110.40 (264.2)
38X	73.50 (1866.9)	36.00 (914.4)	10.40 (264.2)
50X	90.00 (2286.0)	36.00 (914.4)	10.40 (264.2)
38X	73.50 (1866.9)	44.00 (1117.6)	10.40 (264.2)
50X	90.00 (2286.0)	44.00 (1117.6)	10.40 (264.2)

① Box depth is 10.40 inches (264.2 mm), cover adds 0.90 inches (22.9 mm) for overall enclosure depth of 11.30 inches (287.0 mm).
② 800 A maximum bus size in 24.00 inches (609.6 mm) wide box.

Note: Flush trims available on PRL4X panels with Door-in-Door enclosure only.

Top and Bottom Gutters (Minimum)

- 10.62 inches (269.9 mm)

Side Gutters (Minimum)

- 24.00-inch (609.6 mm) wide box: 5.00 inches (127.0 mm)
- 36.00-inch (914.4 mm) wide box: 6.00 inches (152.4 mm)
- 44.00-inch (1117.6 mm) wide box: 8.00 inches (203.2 mm)

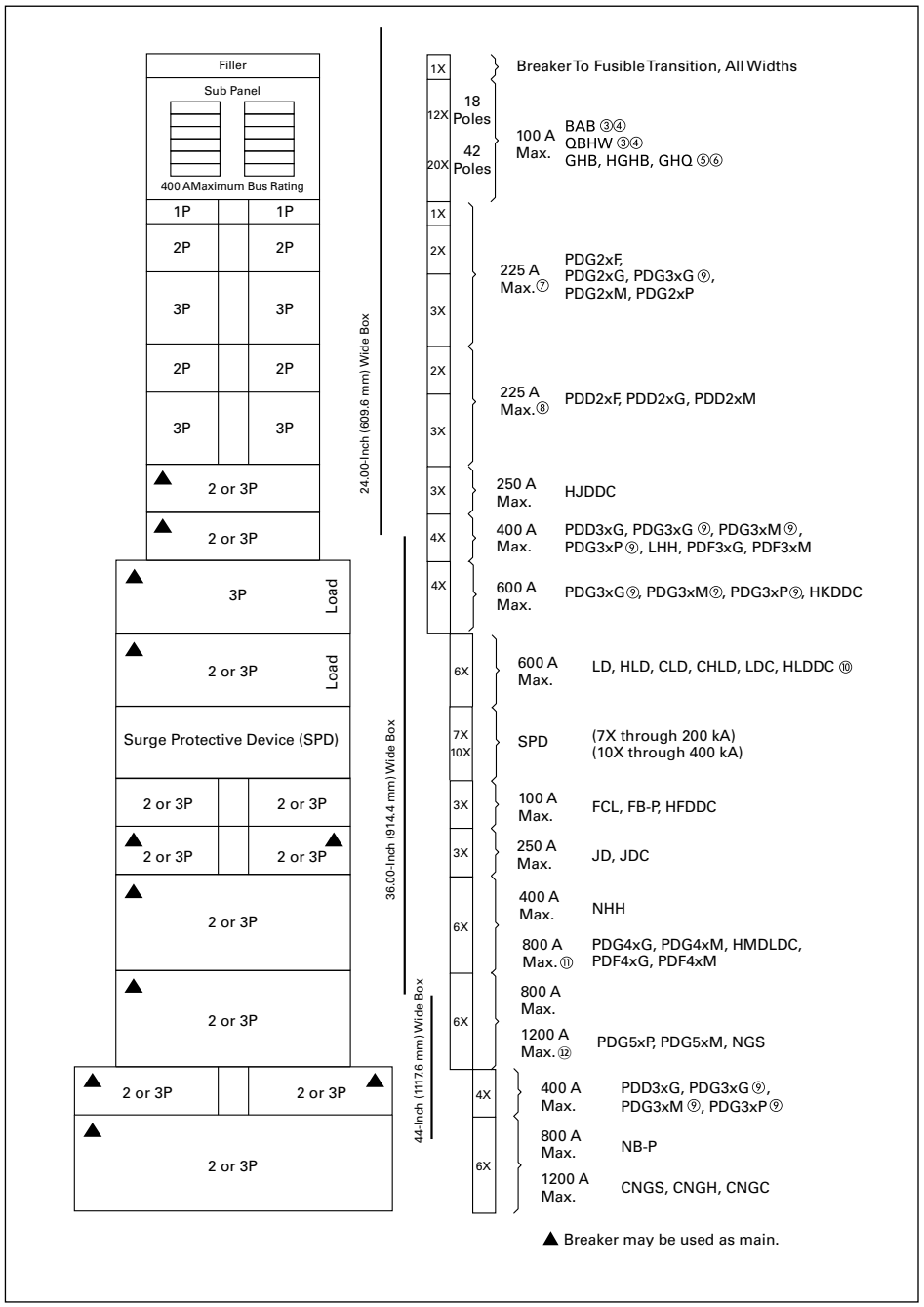


Figure 22.3-6. Layout for Branch and Horizontally Mounted Main Devices

③ BAB and QBHW breakers with shunt trips require one additional pole space, i.e., single-pole is two-pole size, two-pole is three-pole size, and three-pole is four-pole size.
④ If panel contains only BAB or QBHW branch breakers, use a PRL1X panelboard.
⑤ GHB, HGHB or GHQ breakers cannot be mixed on same subchassis as BAB, QBHW.
⑥ If panel contains only GHB, HGHB or GHQ branch breakers, use a PRL2X panelboard.
⑦ When only one single-pole breaker of the group is required on either side of chassis, the single-pole breaker space required changes from 1X to 2X.
⑧ Minimum 36.00-inch (914.4 mm) wide box is required if optional #6–300 kcmil lug is required.
⑨ The 400 A frame must use trip units of ratings 100–400, while the 600 A frame must use trip units of ratings 500, 600 or designated by H, such as H250. The H as the leading character of the ampacity indicates a high instantaneous version of the breaker for coordination purposes. H ratings must use 600 A frame.
⑩ For use on DC systems only.
⑪ MDL main breaker in 24.00-inch (609.6 mm) wide box, refer to **Figure 22.3-4**.
⑫ Optional 750 kcmil terminal requires 44.00-inch (1117.6 mm) wide box.

Note: See **Page 22.3-20** for MLO or Neutral and Vertically Mounted Mains space requirements.

PRL4DX Layout Guide

Technical Data

Instructions

Determine box size by locating all main and feeder devices in your panel. The width of box is determined by the maximum box size shown for each device. For main lugs, through-feed lugs and sub-feed lugs.

Table 22.3-13. Standard Panelboard and Box —Dimensions in Inches (mm)

Panel Height	Box Height	Box Width	Box Depth ①
38X	73.50 (1866.9)	36.00 (914.4)	11.31 (287.0)
50X	90.00 (2286.0)	36.00 (914.4)	11.31 (287.0)
38X	73.50 (1866.9)	44.00 (1117.6)	11.31 (287.0)
50X	90.00 (2286.0)	44.00 (1117.6)	11.31 (287.0)

① Box depth is 10.40 inches (264.2 mm); cover adds 0.90 inches (22.9 mm), for overall enclosure depth of 11.30 inches (287.0 mm).

Note: Flush trims not available on PRL4DX panels. Door-to-door option not available on PRL4D panels.

Top and Bottom Gutters (minimum)

- 10.63 inches (269.9 mm)

Side Gutters (minimum)

- 36.00-inch (914.4) wide box:
  - 6.00 inches (152.4 mm)
- 44.00-inch (1117.6 mm) wide box:
  - 8.00 inches (203.2 mm)

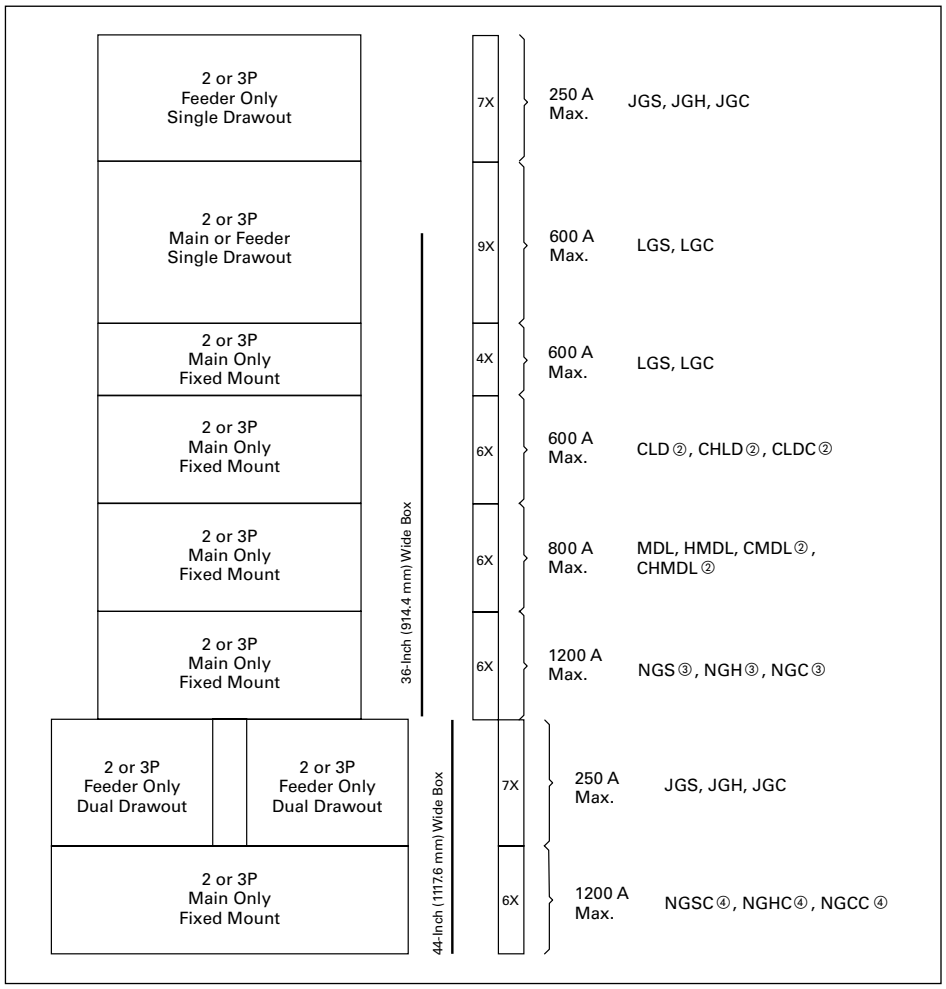


Figure 22.3-7. Layout for Branch and Horizontally Mounted Main Devices—PRL4D

② 100% rated breaker.
③ Optional 750 kcmil terminal requires 44.00-inch (1117.6 mm) wide box.
④ Contact Eaton for availability.

Series Rated Combinations

UL permits panelboards to be labeled with a short-circuit rating of up to 200,000 A symmetrical where UL listed combinations of main and branch circuits are used.

These combinations consist of main breakers or fusible devices connected ahead of, and in series with approved conventional devices.

Two arrangements are acceptable and comply with UL standards for panelboards. The main circuit breaker may be installed in the panel as a main device (**Figure 22.3-8**), or it may be mounted remote (**Figure 22.3-9**) from the panel. In either case, the approved main and branch combinations must be followed. These arrangements are acceptable and are UL listed having been tested in accordance with UL standards.

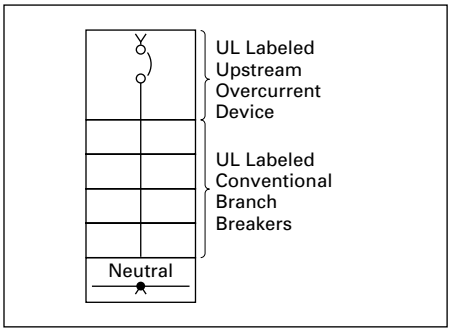


Figure 22.3-8. Main Device

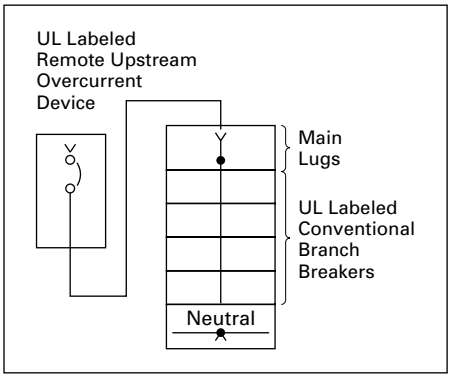


Figure 22.3-9. Mounted Remote

Industry standards and the NEC require protection of the entire electrical distribution system from damage due to short-circuit faults. Article 230.205 of the NEC states that service entrance equipment shall be suitable for the short-circuit current available at its supply terminals. The entire distribution system is required to meet this standard. Series rated systems have become an effective method of meeting these requirements.

There are three protection systems used to protect low voltage power distribution equipment. They are:

- Fully rated protection
- Fully rated, selectively coordinated protection
- Series rated protection

Fully Rated Protection—All overcurrent devices are rated for the full prospective short-circuit current at their line side terminals throughout the system.

Selectively Coordinated Protection—A fully rated system where the over-current device closest to the fault will open first, thus isolating the faulty circuit.

Series Rated Protection—A short-circuit interrupting rating assigned to a combination of two or more over-current protective devices that are connected in series and in which the rating of the downstream device(s) in the combination is less than the series rating.

Series ratings are also known in the industry as integrated ratings, series combination ratings and series connected ratings.

UL Issues

In a series rated system, all of the overcurrent devices in series in the protective scheme must have been tested and listed by Underwriters Laboratories for series combination use in the system.

All Eaton's series ratings are in full compliance with all applicable requirements of the latest editions of UL 489, 891 and 67.

The UL Recognized Component Directory (the Yellow Book) contains breaker manufacturers' series connected listings. These are intended ONLY as a guideline for use by others who are responsible for their own testing, labeling and listing. Therefore, the UL Recognized Component Directory cannot be used to interpret series connected ratings in assembled equipment. The assembled equipment must also be UL tested for series ratings.

Code Issues

The fault current contribution of motors connected between series rated breakers must be considered. Article 240.86 in the NEC states that for series ratings, the sum of the motor full-load currents cannot exceed 1% of the interrupting rating of the lower rated circuit breaker. The actual fault current contribution from induction motors is about four times their full-load current (impedance value of 25%). For example, if the downstream branch circuit breakers used in a series rated combination have an interrupting rating of 14,000 A rms symmetrical for a 480 V system, the maximum allowable motor contribution to that panel from the branch circuit breakers is 140 A (1%). For typical induction motors, this is equivalent to a total horsepower at 480 V of approximately 115 horsepower.

Requirements of the NEC (NFPA 70®) for series ratings may be met by equipment marked with ratings adequate for the available fault current at the point of application in the electrical system. Eaton panelboards and switchboards are marked consistent with NEC Article 240.83.

Additionally, Article 110.22 requires field marking on equipment where series ratings are used. This label is supplied standard with all Eaton panelboards and switchboards.



NEC Required Caution Label

Note: The NEC requires the installer to properly apply and complete this label. Label(s) must be placed on all equipment where series ratings are used.

Fuse Application Considerations

Fuses can be used instead of circuit breakers in fully rated, selectively coordinated and series connected protection systems.

Don't apply fuses using the up-over-down method, which has been recommended by some fuse manufacturers for sizing a current-limiting fuse that protects a downstream molded case circuit breaker with a specified rms symmetrical interrupting rating. The method can lead to erroneous and unsafe conclusions, and should not be used.

Example: Assume a specific type of current-limiting fuse rated 2000 A. Then using the figure below:

1. Draw a vertical line from the prospective short-circuit current of 200 kA to intersect the typical peak let-through curve at "A."
2. Draw a horizontal line left from Point "A" to intersect the "prospective peak" curve at "B."
3. Drop a vertical line from "B" to intersect the horizontal axis and read the recommended rating, 65 kA rms, concluding that a circuit breaker with a 65 kA interrupting capacity will be protected by a specified 2000 A current-limiting fuse.

This conclusion is wrong when the downstream service has a blow-open contact assembly, as does a molded case circuit breaker or similar device.

The reason: The up-over-down method ignores dynamic impedance (the inherent current limiting of the downstream molded case circuit breaker). Such impedance is developed directly by the forces of the let-through current created when the contacts are blown open.

For proper application of current-limiting fuses, always refer to recommendations by the manufacturer of the circuit breaker, which are based on actual test data.

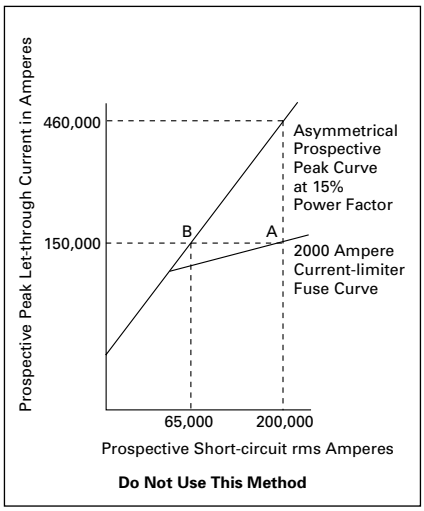


Figure 22.3-10. Old Up-Over-Down Chart

Applications of Series Ratings

Series ratings can also be applied under the following guidelines:

Any FULLY RATED breaker can be applied upstream, downstream or in the middle of any of the series ratings stated in the tables.

Any series rating stated in the tables may have additional branch breakers of the EXACT SAME TYPE further downstream in that rating.

COMBINING SERIES RATINGS is allowed under certain conditions. Main and branch series ratings may be combined if:

- Breakers A, B and C are in series respectively from main to branch
- Breakers A and B series rate together
- Breakers A and C series rate at the same interrupting rating level (or higher)
- It is allowable to use A, B and C together at the A-B series rating

It is improper to combine series ratings under the following condition:

- Breakers A, B and C are in series respectively from main to branch
- Breakers A and B series rate together
- Breakers B and C series rate at the breaker B interrupting rating level
- It is NOT allowable to use A, B and C together at the A-B series rating

However, combining multiple overcurrent devices as in this example, can be accomplished if all devices in the series combination have been tested together.

Series Rating Combinations

For more information on available series rating combinations, please see Information Manual 1C96944H02.

Panelboard Selection Guide

Table 22.3-14. Product Types

			
Type PRL1X	**Fusible Lighting Panelboard PRL1XF**	**Type PRL1X-LX Column Type**	**Type PRL2X**
Bolt-on or plug-on circuit breakers 240 Vac maximum	240 Vac maximum	Bolt-on circuit breakers 240 Vac maximum	Bolt-on or plug-on circuit breakers 240 or 480Y/277 Vac 125/250 Vdc maximum
Main lugs only 600 A maximum	Main lugs only 400 A maximum	Main lugs only 225 A maximum	Main lugs only 600 A maximum
Main circuit breaker 600 A maximum		Main circuit breaker 225 A maximum	Main circuit breaker 600 A maximum
Branch circuit breakers 100 A maximum Single, two- and three-pole	Branch overcurrent protective devices 30 A maximum Single-, two- and three-pole utilizing Class CC fuses	Branch circuit breakers 100 A maximum Single-, two- and three-pole	Branch circuit breakers 100 A maximum Single-, two- and three-pole

		
Fusible Lighting Panelboard PRL2XF	**Type PRL2X-LX Column Type**	**Retrofit Panelboard PRL-1RX and PRL-2RX**
240 and 480Y/277 Vac maximum	Bolt-on or plug-on circuit breakers 240 or 480Y/277 Vac 125/250 Vdc maximum	Bolt-on or plug-on circuit breakers 240 or 480Y/277 Vac
Main lugs only 400 A maximum	Main lugs only 225 A maximum	Main lugs only 225 A maximum
	Main circuit breaker 225 A maximum	Main circuit breaker 225 A maximum
Branch overcurrent protective devices 30 A maximum Single-, two- and three-pole utilizing Class CC fuses	Branch circuit breakers 100 A maximum Single-, two- and three-pole	Branch circuit breakers 100 A maximum Single-, two- and three-pole

Table 22.3-14. Product Types (Continued)

			
Type PRL3E	**Type PRL3X**	**Type PRL3XF**	**Type PRL4X / PRL4F**
Bolt-on circuit breakers 240 Vac maximum	Bolt-on circuit breakers 240, 480 or 600 Vac; 250 Vdc maximum	Bolt-on fusible switches 600 Vac; 125 Vdc maximum	Circuit breakers or fusible switches; 240, 480 or 600 Vac; 600 Vdc maximum
Main lugs only 600 A maximum	Main lugs only 800 A maximum	Main lugs only 400 A maximum	Main lugs only 1200 A maximum
			Main circuit breaker 1200 A maximum
Main circuit breaker 600 A maximum	Main circuit breaker 600 A maximum	Main fused or unfused switch 400 A maximum	Main fusible switch 1200 A maximum
			Branch circuit breakers 1200 A maximum, Single-, two- and three-pole
Branch circuit breakers 125 A maximum Single-, two- and three-pole	Branch circuit breakers 225 A maximum Single-, two- and three-pole	Branch Bussmann Compact Circuit Protector Base with CUBEFuse 100 A maximum Single-, two- and three-pole	Branch fusible switches 1200 A maximum, Single-, two- and three-pole

		
Pow-R-Command	**Multipoint Metering Distribution Panelboard**	**Elevator Control Panelboard**
Bolt-on or plug-on circuit breakers 240 or 480Y/277 Vac	Bolt-on circuit breakers 600 Vac or 600 Vdc maximum	Bolt-on fusible switches 600 Vac maximum
Main lugs only 400 A maximum	Type PRL4X panelboard specially formatted to provide a compact and flexible multipoint metering solution for 250–1200 A applications	Controls for up to four elevators in a single panelboard
Main circuit breaker 400 A maximum		Main lugs only 800 A maximum
Branch circuit breakers 225 A maximum Single-, two- and three-pole		Branch overcurrent devices 15–200 A fusible switches with Class J fuse clips maximum
Single- and two-pole remote operated circuit breakers		
Integral load switching and dimming controls		Designed to meet specific sections various codes impacting elevators

Table 22.3-15. Panelboard Selection Guide

Panelboard Type	Device Type	Maximum Voltage Rating		Maximum Main Rating, Amperes		Branch Circuits Ampere Range	Short-Circuit Current Ratings rms Symmetrical Amperes, AC	
		AC	DC	Main Lugs Only	Main Device		Fully Rated (kA)	Series Rated (kA)
Pow-R-Line 1X ①	Breaker	240	—	600	600	15–100	10–22	22–200
Pow-R-Line 2X ①	Breaker	240 480Y/277	250	600 400 ②	600 400 ②	15–100 15–100	65 14	65–200 22–150
Pow-R-Line 3X ①	Breaker	240 480 600	250	800 800 800	600 600 600	15–225 15–225 15–225	10–200 14–100 14–35	22–200 22–150 —
Pow-R-Line 3E ①	Breaker	480	250	600	600	15–125	35–65	35–100
Pow-R-Line 4X ①	Breaker	240 480 600	600	1200 1200 1200	1200 1200 1200	15–1200 15–1200 15–1200	10–200 14–200 14–200	22–200 22–150 —
Pow-R-Line 4F ①	Fusible switch	240 600	250	1200 1200	1200 1200	30–1200 30–1200	100–200 100–200	— —
Pow-R-Line 4DX	Breaker	240 480	— —	1200 1200	1200 1200	20–600 Drawout 15–1200 Fixed	100 65	— —
Pow-R-Line 1XF	Fusible switch	240	—	400	400	15–30	200	200
Pow-R-Line 2XF	Fusible switch	480Y/277	—	400	400	15–30	200	200
Pow-R-Line 3XF	Fusible switch	600	125	400	400	15–100	200	200
Pow-R-Line 1X-LX	Breaker	240	—	225	225	15-30	10–22	18–200
Pow-R-Line 2X-LX	Breaker	480Y/277	125/250	225	225	15–30	14	25–150
Pow-R-Line 1RX	Breaker	240	—	225	225	15–100	10–22	22–100
Pow-R-Line 2RX	Breaker	480Y/277	—	225	225	15–100	14	22–150
Elevator control panelboard ①	Fusible	480	—	800	800	15–200	10–200	14–100

① Available with surge protective device (SPD) and metering.
② Amperage rating for DC voltage.

Type PRL4X Modifications



Type PRL4X

Modifications Selection Guide

Table 22.3-16. Modifications—Alphabetical Index

Modification	Available on Panelboard Types
	PRL4X
Ambient compensating breakers	Yes
Bus density	Yes
Cabinets—special: Types 2, 3R, 4, 4X, 12	Yes
Complete assembly	Yes
Compression type lugs, mains only	Yes
Concealed trim clamps (LT trim)	No
Conduit covers	Yes
Copper lugs	Yes
Copper main bus	Yes
Directory frame—metal	Yes
Doors, special	Yes
Electronic trip units	Yes
Fungus-proof	Yes
Ground bar	Yes
Ground fault protection (zero sequence)	Yes
Handle lockoff device	Yes
Hinges, special (LT trim)	Yes
Increased dimensions	No
Increased panel bus rating	No
Interiors to fit existing boxes	Yes
Locks, special (LT trim)	Yes
Metering devices	Yes
Molded case switches	Yes
Nameplates engraved	Yes
Neutral rated 200%	Yes
Painting and special coating	Yes
Permanent circuit numbers	Yes
Remote control switches (ASCO 920)	No
Service entrance	Yes
Shunt trips	Yes
Split bus or meter loop	No
Sub-feed breakers	No
Sub-feed lugs	Yes
Sub-metering	Yes
Surge protective device (SPD)	Yes
Tamperproof screws (LT trim)	Yes
Terminals, copper only for breakers	Yes
Through-feed lugs	Yes
Time clock space only	—
Touchup paint	Yes

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